

Problem Analysis Notes

1. Weld Defect Prediction

- **Task:** Identify weld defects.
 - **ML Fit:** High value — no single rule reliably predicts defects from process parameters alone.
 - **Impact:** High — missed structural weld defects can lead to catastrophic failure.
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2. HVAC Compressor Failure (Vibration Data)

- **Task:** Estimate remaining days or hours until failure using continuous sensor data.
 - **ML Fit:** High value — degradation patterns over time suit machine learning better than fixed maintenance intervals.
 - **Impact:** Moderate — incorrect predictions cause unplanned outages or wasted maintenance spend.
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3. Soil Type Classification (Borehole Logs)

- **Task:** Classify soil type (clay, silt, sand, gravel, fill).
 - **ML Fit:** Medium value — standardizes human judgment calls across photos and field notes.
 - **Impact:** Low to Moderate — errors are typically caught downstream by lab testing.
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4. Bridge Legal Load Rating

- **Task:** Determine legal load ratings for existing bridges.
- **ML Fit:** Not Suitable — governed strictly by structural calculations and legal codes; statistical models carry no legal standing.
- **Impact:** Extreme risk — incorrect ratings risk structural collapse and loss of life.